

Week 5: Key Concepts and Terms

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This deck collects the **key concepts and terms** from Week 5: Panel Regression Recap.

For each term you should be able to:

- Define it in your own words
- Give an example or illustration
- Explain why it matters for your dissertation workflow

Use these slides as prompts — prepare your own definitions, answers and materials.

Note: This is a study aid only. It does not attempt to be comprehensive and should not be relied upon as sufficient preparation for assessment purposes.



Panel Data Structure

What Is Panel Data?

Term	What to know
Panel data	What are its two dimensions? How does it differ from pure cross-sectional or pure time-series data?
Cross-section	What does the cross-sectional dimension represent (e.g. firms, countries)?
Time series	What does the time dimension represent (e.g. years, months)?
Unit–time pair	What is a “firm–year” observation? Why is each row identified by two things?



Term	What to know
Unobserved heterogeneity	What are examples of things you cannot directly measure (e.g. firm culture, management quality)?
Within-unit variation	What does it mean to compare a firm “to itself” over time?
Between-unit variation	What does it mean to compare different firms to each other? Why can this be misleading?

The Pooled OLS Problem

Term	What to know
Pooled OLS	What does it mean to run OLS on panel data without fixed effects? What assumption does it make?
Omitted variable bias (OVB)	What happens to $\hat{\beta}$ when an unobserved variable correlates with both X and Y ?
Confounding variable	What is a confounder? How does it create a spurious relationship between X and Y ?
Endogeneity	What does it mean for X to be correlated with the error term ε ?

Fixed Effects

Term	What to know
Fixed effect (α_i)	What is a unit-specific intercept? What does it absorb?
Demeaning	What does it mean to subtract each unit's time average? Why is this equivalent to including α_i ?
Time-invariant	What kinds of variables are “absorbed” by firm fixed effects (e.g. founding year, geography)?
Singleton observation	What happens to firms with only one observation? Why are they dropped?



Term	What to know
Firm FE (α_i)	What time-invariant firm traits does it absorb?
Year FE (δ_t)	What common time shocks does it absorb (e.g. financial crises, regulatory changes)?
Two-way FE	What does including both firm and year FE simultaneously control for?
Industry $\times$ Year FE	What does an interacted fixed effect absorb that separate FEs cannot?

Term	What to know
No dynamics assumption	Why does FE require that past outcomes do not cause current treatment? (Imai & Kim 2019)
Within variation only	Why does $\hat{\beta}$ come only from within-unit changes, not cross-sectional differences?
Plausible counterfactuals	Why should you use within-unit SD (not overall SD) for effect-size claims? (Mummolo & Peterson 2018)
Residualizing X	What does it mean to “residualize” X with respect to your fixed effects, and why does it reveal the variation driving $\hat{\beta}$?



Clustered Standard Errors

Why OLS Standard Errors Fail in Panels

Term	What to know
Independence assumption	What does OLS assume about the correlation between ε_{it} and ε_{js} ? Why does this fail in panels?
Firm effect (serial correlation)	Why are residuals for the same firm correlated across years?
Time effect (cross-sectional correlation)	Why are residuals for different firms correlated within the same year?
Over-rejection	What happens to hypothesis tests when SEs are too small? (Petersen 2009: 15% reject at 1%)



Term	What to know
Clustered standard errors	What assumption do they relax? How do they allow correlation within clusters?
Cluster by firm	When should you cluster by firm? What dependence structure does it handle?
Cluster by time	When should you cluster by time? When might you prefer Fama–MacBeth?
Two-way clustering	What does $\hat{V}_{\text{firm}} + \hat{V}_{\text{time}} - \hat{V}_{\text{White}}$ achieve? When do you need it?
Minimum cluster count	Why do you need at least 30–50 clusters for reliable inference?

Term	What to know
Petersen's diagnostic	How does comparing SE methods (IID, robust, firm-clustered, time-clustered) reveal the dependence structure?
Treatment assignment level	Why should you cluster at (or above) the level at which treatment varies?
Over-clustering	What do Abadie et al. (2023) show about unnecessary clustering destroying statistical power?
Robust (HC1) SEs	When might heteroskedasticity-robust SEs suffice instead of clustering?

Running Regressions with `pyfixest`

Term	What to know
<code>pf.feols()</code>	What are its three key arguments (formula, vcov, data)?
Formula: <code>y ~ x1 + x2</code>	What does the left-hand side represent? What does + mean on the right?
The <code> </code> separator	What goes after the pipe? How does it differ from Stata's <code>absorb()</code> ?
<code>vcov</code> argument	What are the options: <code>"iid"</code> , <code>"HC1"</code> , <code>{"CRV1": "firm"}</code> , <code>{"CRV1": "firm + year"}</code> ?

Term	What to know
<code>.coef()</code> , <code>.se()</code> , <code>.pvalue()</code>	What does each method return from a fitted model?
<code>.tidy()</code>	What does it return and why is it useful for further analysis?
<code>.confint()</code>	What is a confidence interval and how do you extract one?
<code>.vcov()</code> after estimation	Why is it useful to re-compute SEs without re-running the regression? How does this implement Petersen's diagnostic?
<code>csw0()</code>	What does cumulative stepwise estimation do? Why run models that progressively add controls?

Presenting Results

Term	What to know
<code>pf.etable()</code>	How does it display multiple models side by side? What arguments does it take?
<code>labels</code>	Why should you replace variable names with human-readable labels?
<code>type="tex"</code>	How do you export a table as $\text{\LaTeX}$ for your dissertation?
<code>coef_fmt</code>	How do you control what appears below the coefficient (SEs vs t -stats)?

Term	What to know
<code>pf.coefplot()</code>	What does a coefficient plot show? When is it more effective than a table?
Confidence interval	What does the interval around each coefficient represent?
<code>hypotheses()</code> (marginaleffects)	How do you test whether two coefficients are equal?
Delta method	What does it compute and why is it needed for non-linear functions of coefficients (e.g. ratios)?

Key Papers

Four Papers You Should Know

Paper	Key insight
Petersen (2009)	Compare SE methods to diagnose the dependence structure of your data; firm effect $\rightarrow$ cluster by firm.
Imai & Kim (2019)	FE assumes no dynamic feedback between treatment and outcome; violations make $\hat{\beta}$ inconsistent.
Mummolo & Peterson (2018)	Use within-unit SD for counterfactuals; many published effect sizes are overstated.
Abadie et al. (2023)	Cluster at the level of treatment assignment; unnecessary clustering wastes power.



Summary Checklist

Can you define and give an example for each?

- Panel data
- Cross-section / time series
- Unit-time pair
- Pooled OLS
- Omitted variable bias
- Endogeneity
- Fixed effect (α_i)
- Demeaning
- Firm / year / two-way FE
- Industry $\times$ year FE
- Singleton observation
- Within-unit variation
- Clustered standard errors
- Two-way clustering
- Petersen's diagnostic
- `pf.feols()` / the `|` separator
- `pf.etable()` / `pf.coefplot()`
- Delta method



Can you answer these in your own words?

- 1 What is panel data and how does it differ from cross-sectional data?
- 2 What is omitted variable bias and why does pooled OLS suffer from it?
- 3 What does a firm fixed effect absorb, and what can it *not* absorb?
- 4 Why did the patent stock coefficient drop from 0.847 to 0.486 when firm FE were added?
- 5 What is the trade-off when adding more fixed effects to a model?
- 6 Why are OLS standard errors biased in panel data?
- 7 When should you cluster by firm vs. by time vs. two-way?
- 8 In `pyfixest`, what does the `|` in the formula `y ~ x | firm + year` do?

